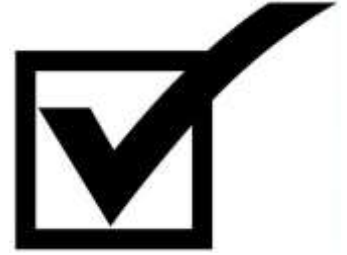




**BUSINESS
PROFESSIONALS
of AMERICA**
Giving Purpose to Potential



MANAGERIAL ACCOUNTING (135) STATE 2023

CONCEPT KNOWLEDGE:

Multiple Choice (20 @ 2 points each) _____ (40 points)

APPLICATION KNOWLEDGE:

Short Answer _____ (75 points)

Problem 1: CVP Analysis _____ (30 points)

Problem 2: Profitability Analysis _____ (30 points)

Problem 3: Process Costing _____ (30 points)

TOTAL POINTS _____ **(205 points)**

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

EXAM GUIDELINES:

You have been hired as a Financial Assistant and will be keeping the accounting records for Digital Solutions, located at 700 Morse Road, Suite 201, Columbus, Ohio 43214. Digital Solutions provides accounting and other financial services for clients. You will complete jobs for Digital Solutions' own accounting records, as well as for clients.

You will have 90 minutes to complete your work. The test is divided into two parts: concept knowledge and application knowledge.

Your name and/or school name should *not* appear on any work you submit for grading. Write your Member ID in the provided space. Staple all pages in order before you turn in your test.

Assumptions to make when taking this assessment:

- Round all calculations to two decimal places at the final step.
- Round all percentages to one decimal place.
- Use 360 days for interest calculations.

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

- | | | | |
|-----|---|-----|---|
| 1. | A | 11. | C |
| 2. | D | 12. | A |
| 3. | D | 13. | B |
| 4. | B | 14. | A |
| 5. | A | 15. | D |
| 6. | D | 16. | B |
| 7. | C | 17. | D |
| 8. | B | 18. | A |
| 9. | C | 19. | A |
| 10. | C | 20. | A |

Short Answer (25 @ 3 Points Each = 75 Points Total)

1. McGonagall Company sells only two products, Product A and Product B. McGonagall sells two units of Product A for each unit it sells of Product B.

	Product A	Product B	Total
Selling price	\$50	\$20	
Variable cost per unit	\$17	\$4	
Total fixed costs			\$2,132,000

How many units of Product A does McGonagall need to sell to break-even?	52,000
How many units of Product B does McGonagall need to sell to break-even?	26,000

2. Potter Company has the following data for the Assembly Department for August:

Units in process at the beginning of August	700
Units started in August	2,200
Units completed and transferred	?
Units in process at end of August	800

Conversion costs are added evenly throughout the process. The company uses the weighted-average method. Compute the equivalent units of production for direct materials and conversion costs for each independent scenario:

	Direct Materials	Conversion Costs
Units in process at the end of August are 20% complete; materials are added at the beginning of the process.	2,900	2,260
Units in process at the end of August are 80% complete; materials are added at the beginning of the process.	2,900	2,740
Units in process at the end of August are 45% complete; materials are added at the end of the process.	0	2,460
Units in process at the end of August are 78% complete; materials are added at the halfway point.	2,900	2,724

3. HPU is an American beverage company that manufactures energy drinks. The company calculates product cost per pallet of finished and packaged product. For each pallet, \$3,100 of direct material is used and there is \$2,500 of direct manufacturing labor at \$25 per hour. Manufacturing overhead is applied at \$30 per direct manufacturing labor hour.

Calculate the profit earned on 45 pallets if each pallet sells for \$10,000.	\$1,400
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4. Sirius, Inc. reports the following information for July:

Sales Revenue	\$1,000,000
Variable Costs	120,000
Operating Income	390,000

What is the total fixed cost using variable costing?	\$490,000
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5. Carrey Company manufactures a single product. Annual production costs incurred in the manufacturing process are shown below for the production of 2,000 units. The utilities and maintenance are mixed costs; the fixed portion of these costs are \$200 and \$400 respectively. Complete the chart below, determining the total costs for the production of 4,000 units

	Costs Incurred	
Production in Units:	2,000	4,000
Production Costs:		
Direct Materials	\$4,000	\$8,000
Direct Labor	16,000	32,000
Utilities	1,000	1,800
Rent	3,000	3,000
Indirect Labor	4,600	9,200
Supervisor Salaries	1,500	1,500
Maintenance	900	1,400
Depreciation (straight-line)	2,500	2,500

6. Daniel is considering opening a Kwik Oil Change Center. He estimates that the following costs will be incurred during his first year of operations: Rent \$6,000, Depreciation on equipment \$7,000, Wages \$16,400, Motor oil \$1.80 per quart. He estimates that each oil change will require 5 quarts of oil. Oil filters will cost \$3.00 each. Daniel must also pay The Kwik Corporation a franchise fee of \$1.40 per oil change, since he will operate his business as a franchise. In addition, utility costs are expected to behave in relation to the number of oil changes as follows:

Number of Oil Changes	Utility Costs
4,000	\$6,000
6,000	7,300
19,000	15,000
12,000	12,600
9,400	10,200

Daniel anticipates that he can provide the oil change service with a filter for \$25.00 each.

Using the high-low method, determine variable cost per unit for utilities.	\$0.60
Using the high-low method, determine fixed costs for utilities.	\$3,600
Determine the break-even point in number of oil changes	3,000
Determine the break-even point in sales dollars	\$75.,000
Determine the number of oil changes required to earn net income of \$20,000.	4,819

Problem 1 (30 Points – 5 Points for Each Answer)

Little Whinging Company sells only two products, Product A and Product B.

	Product A	Product B	Total
Selling price	\$40	\$50	
Variable cost per unit	\$24	\$40	
Total fixed costs			\$840,000

Little Whinging sells two units of Product A for each unit it sells of Product B. Little Whinging faces a tax rate of 30%.

How many units of Product A does Little Whinging need to see to break-even?	40,000
How many units of Product B does Little Whinging need to see to break-even?	20,000
How many units of Product A would need to be sold if Little Whinging desired an after-tax net income of \$73,500?	45,000
How many units of Product B would need to be sold if Little Whinging desired an after-tax net income of \$73,500?	22,500
If Little Whinging's tax rate is reduced to 25%, how many units of Product A would need to be sold to earn after-tax net income of \$73,500?	44,667
If Little Whinging's tax rate is reduced to 25%, how many units of Product B would need to be sold to earn after-tax net income of \$73,500?	22,334



Problem 2 (30 Points – 5 Points for Each Answer)

Rubeus Co. manufactures and sells adjustable windows for remodeling homes and new housing. Rubeus developed its budget for the current year assuming that the windows would sell at a price of \$500 each. The variable costs for each window were forecasted to be \$250 and the annual fixed costs were forecasted to be \$130,000. Rubeus had targeted a profit of \$450,000.

While Rubeus's sales usually increase during the second quarter, the May financial statements reported that sales were not meeting expectations. For the first five months of the year, only 400 units had been sold at the established price, with variable cost as planned, and it was clear that the target profit for the year would not be reached unless some actions were taken. Rubeus's president assigned a management committee to analyze the situation and develop several alternative courses of action. The following three alternatives were presented to the president, only one of which can be selected.

- Alternative 1: Reduce the selling price by \$50. The marketing department forecasts that with the lower price, 3,200 units could be sold during the remainder of the year.
- Alternative 2: Lower variable costs per unit by \$30 through the use of less expensive materials. Because of the difference in materials, the selling price would have to be lowered by \$40 and sales of 2,600 units for the remainder of the year are forecast.
- Alternative 3: Cut fixed costs by \$15,000 and lower the selling price by 5 percent. Sales of 2,200 units would be expected for the remainder of the year.

If no changes are made to the selling price or cost structure, estimate the number of units that must be sold during the year to break-even.	520
If no changes are made to the selling price or cost structure, estimate the number of units that must be sold during the year to attain the target profit of \$450,000.	2,321
Calculate the profitability of each of the alternatives: • Alternative 1	\$610,000
• Alternative 2	\$594,000
• Alternative 3	\$480,000
Which alternative should Rubeus choose?	Alternative 1

Problem 3 (30 Points – 5 Points for Each Answer)

R. Lupin Company was founded by Remus Lupin and David Thewlis in 1929. Early Lupin fishing rods were made from Bamboo, a classic style at the time. Eventually Lupin made the transition from bamboo to fiberglass to graphite. Today, Lupin builds state-of-the-art boron graphite fishing rods. The company uses a weighted-average process-costing system and has one production department for the production of their graphite rods.

All materials are introduced at the start of manufacturing; in contrast, conversion cost is incurred uniformly throughout production. The company had respective work-in-process inventories on May 1 and May 31 of 62,200 units and 71,000 rods, the latter of which was 50% complete. The production supervisor noted that 102,000 rods were completed during the month.

Costs in the May 1 work-in-process inventory were subdivided as follows: materials, \$41,000; conversion, \$92,000. During May, the company's cost accountant charged production with \$339,600 of material and \$730,000 of conversion.

Determine the number of rods that the company started during May.	110,000
Compute the number of equivalent units with respect to conversion cost.	137,500
Determine the conversion cost per equivalent unit.	\$5.98
Compute the cost of the May 31 work-in-process inventory.	\$368,490
Compute the cost of goods manufactured in May.	\$834,360
What account would have been credited to record the company's completed production?	Work-in-Process